



TECHNISCHE HOCHSCHULE
OSTWESTFALEN-LIPPE
UNIVERSITY OF
APPLIED SCIENCES
AND ARTS

Interactive 3D Campus Environments Using Gaussian Splatting in Unreal Engine

- Real-time Radiance Field Rendering for Interactive Campus Exploration

Team 2:

Jahnavi Kalla

Karthik Bommidi

Nithya Kanakam

Keerthana Kothakapu Adamulla

Serin Sara Samuel

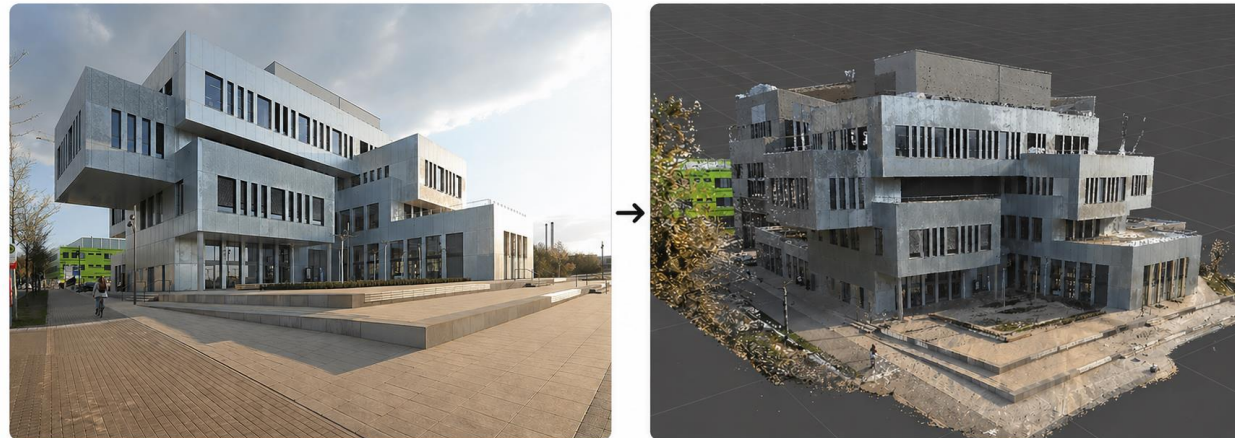
From Campus Images to an Interactive 3D Environment

Problem:

- Real-world campus environments are difficult to convert into realistic, interactive 3D scenes.

Our Goal:

- Reconstruct the TH OWL campus from images and bring it into Unreal Engine for real-time exploration.



Why This Matters?

1. Virtual Campus Access

- Create realistic digital campus spaces for navigation, exploration, and remote access.

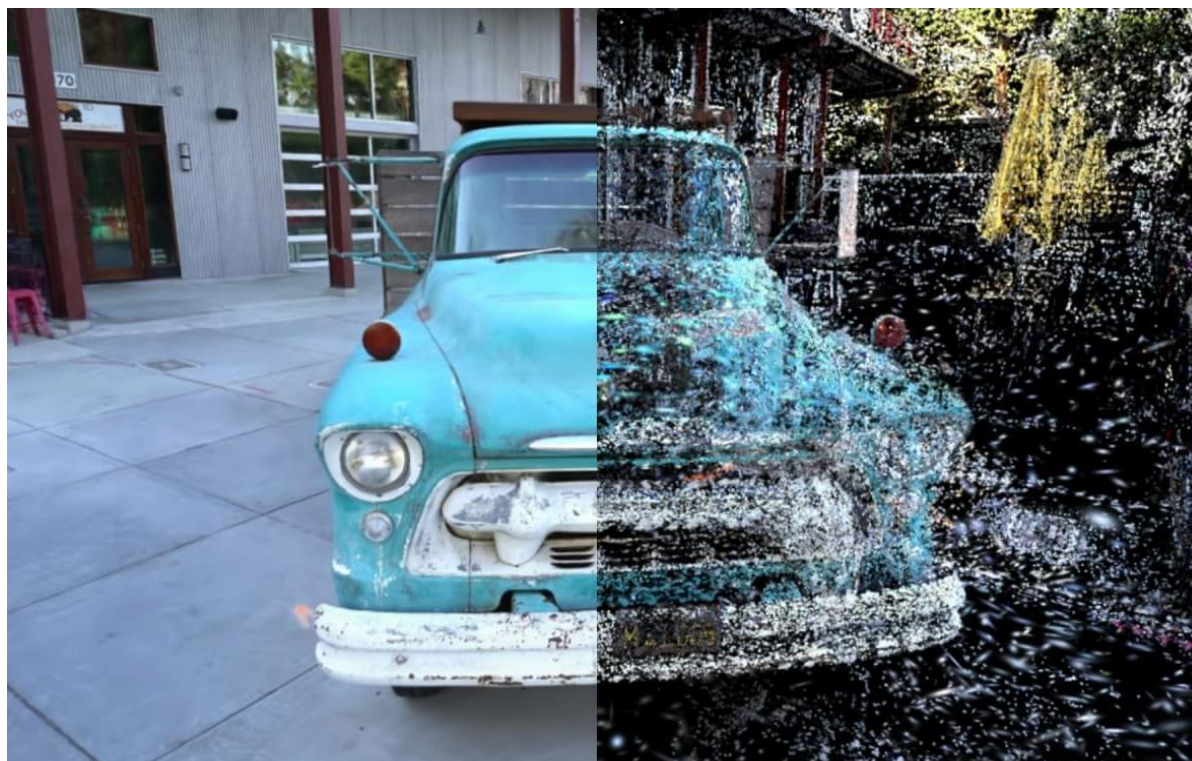
2. Gamified Learning

- Use interactive 3D environments to improve engagement in educational experiences.

3. Digital Twin Foundation

- Build a reusable campus model for future simulation, planning, and smart campus applications.

What is Gaussian Splatting?



A method that turns a scene into many soft 3D ellipsoids and draws them quickly to create a realistic 3D view.

A **Gaussian** is a smooth mathematical function whose shape in 3D forms soft ellipsoids when you look at regions of equal density.

Splatting is projecting each 3D Gaussian onto the 2D screen as an ellipse to form the final image.

Why Gaussian Splatting?

Photogrammetry

- Good geometry, but heavy cleanup and limited realism. Eg: Google maps

NeRF (Neural Radiance Fields)

- High visual quality, but too slow for real-time interaction.

Gaussian Splatting

- Balances realism, cost-effective and real-time performance for interactive 3D scenes.

Gaussian Splatting Workflow

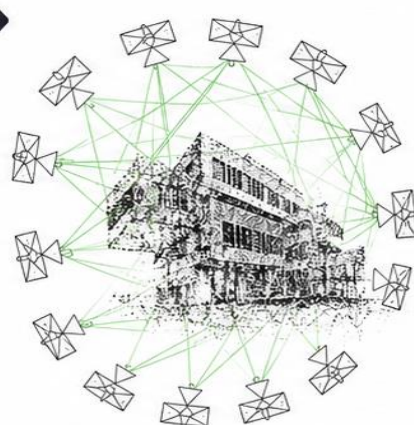
1 Image Capture

Capture overlapping images of the scene from multiple viewpoints.



2 Camera Calibration & Pose Estimation

Estimate camera intrinsics, extrinsics, and scene geometry using Structure-from-Motion.



3 Sparse Point Cloud

Generate a sparse 3D point cloud as the initial scene representation.



4 Gaussian Initialization & Optimization

Convert points into Gaussian splats and optimize their color, opacity, size, and orientation.

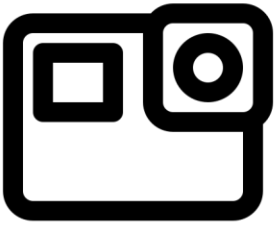


5 Real-Time Rendering

Project optimized splats onto the screen for fast, high-quality rendering.



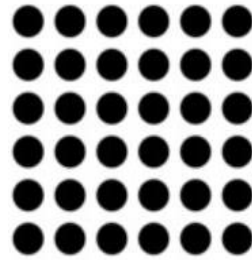
Tools & Technologies Used



Go Pro



Reality Scan



LichtFeld Studio



SuperSplat



Unreal Engine
5.1.1

System Requirements

Local Device

- **Device:** 2023 HP OMEN 16 (Model 16-xf0060AX)
- **Processor & Memory:** AMD Ryzen 7 CPU with 16 GB DDR5 RAM (5600 MHz)
- **Graphics:** NVIDIA GeForce RTX 4060 GPU with 8 GB GDDR6 VRAM

Server Environment

- **Server:** NVIDIA DGX A100 AI Server (TH OWL)
- **Environment:** Docker-based GPU-enabled setup (PyTorch / Jupyter Notebook)
- **Compute Resources:** High-performance multi-GPU system shared among users

Data Collection



- **Hardware:** Manual, ground-level capture using a GoPro camera
- **Scale:** Captured approximately 300 GB of total campus data
- **Targeted Subset:** Subsampled the dataset to 60 GB focusing on the Innovation Spin building to accommodate computational limits.

Implementation

Pipeline 1: Manual

- A GUI-based local workflow focusing on high-fidelity reconstruction using Reality Scan, **LichtFeld Studio** and **SuperSplat**

Pipeline 2: Automated

- A headless, script-based workflow on the **DGX Server**. Optimized for large-scale dataset scalability and server-side processing.

Pipeline 1- Pre-Processing

- **Process Automation**

Automates complex photogrammetry

- **COLMAP Integration**

Uses Structure-from-Motion (SfM) algorithms to handle camera calibration and 3D scene re-construction

- **Spatial Blueprint**

Generates the essential data (cameras.txt, images.txt, points3D.txt) required for Gaussian Splatting

- **VRAM Optimization**

Capped the export resolution to 2,000,000 pixels to prevent hardware crashes during training



REALITYSCAN

Pipeline 1 - Training

Hardware

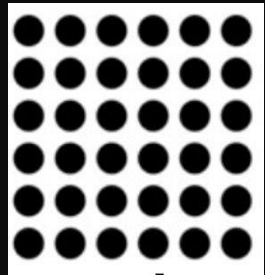
HP OMEN 16 laptop - NVIDIA RTX 4060 GPU - 8 GB VRAM

Training Parameters

- Iterations: 30,000
- SH Degree: 3
- Maximum Gaussians: 2 million

Outcome

Completed training in 14 hours and 37 minutes, utilizing 100% of the available 8.6GB VRAM



LichtFeld **Studio**

Pipeline 2- Server based

- Used **TH OWL NVIDIA DGX** server (A100 GPUs) for high-performance training
- Used Detmold campus dataset (~ 6 GB) to avoid memory issues with 60 GB data
- Accessed via SSH in a headless (no GUI) environment
- Implemented official Graphdeco Gaussian Splatting pipeline (GitHub)
- Developed Bash script to automate preprocessing, training, and export



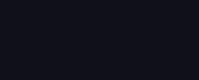
Dataset: Detmold campus

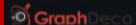
Bernhard Kerbl*, Georgios Kopanas*, Thomas Leimkühler, George Drettakis (* indicates equal contribution)
[Webpage](#) | [Full Paper](#) | [Video](#) | [Other GRAPHDECO Publications](#) | [FUNGRAPH project page](#) |
[T&T+DB COLMAP \(650MB\)](#) | [Pre-trained Models \(14 GB\)](#) | [Viewers for Windows \(60MB\)](#) | [Evaluation Images \(7 GB\)](#)

Method	InstantNGP (9.2 fps)	Plenoxels (8.2 fps)	Mip-NeRF360 (0.071 fps)	Ours (133 fps)	Ours (93 fps)	Ground Truth
Train: 7min, PSNR: 22.1	Train: 26min, PSNR: 21.9	Train: 48h, PSNR: 24.3	Train: 6min, PSNR: 23.6	Train: 51min, PSNR: 25.2		

This repository contains the official authors implementation associated with the paper "3D Gaussian Splatting for Real-Time Radiance Field Rendering", which can be found [here](#). We further provide the reference images used to create the error metrics reported in the paper, as well as recently created, pre-trained models.



Results of Both Pipelines

Metric	Pipeline 1	Pipeline 2
Dataset Size	60 GB	6 GB
Training Mode	GUI-based	Headless
Training Time	14h 37m	~ 45m
Scalability	Limited	High

Super Splat

- Used Super Splat to visualize the Gaussian Splat model
- Edited the splats by removing noise and unwanted artifacts
- Improved model clarity by focusing on the main structure(building)

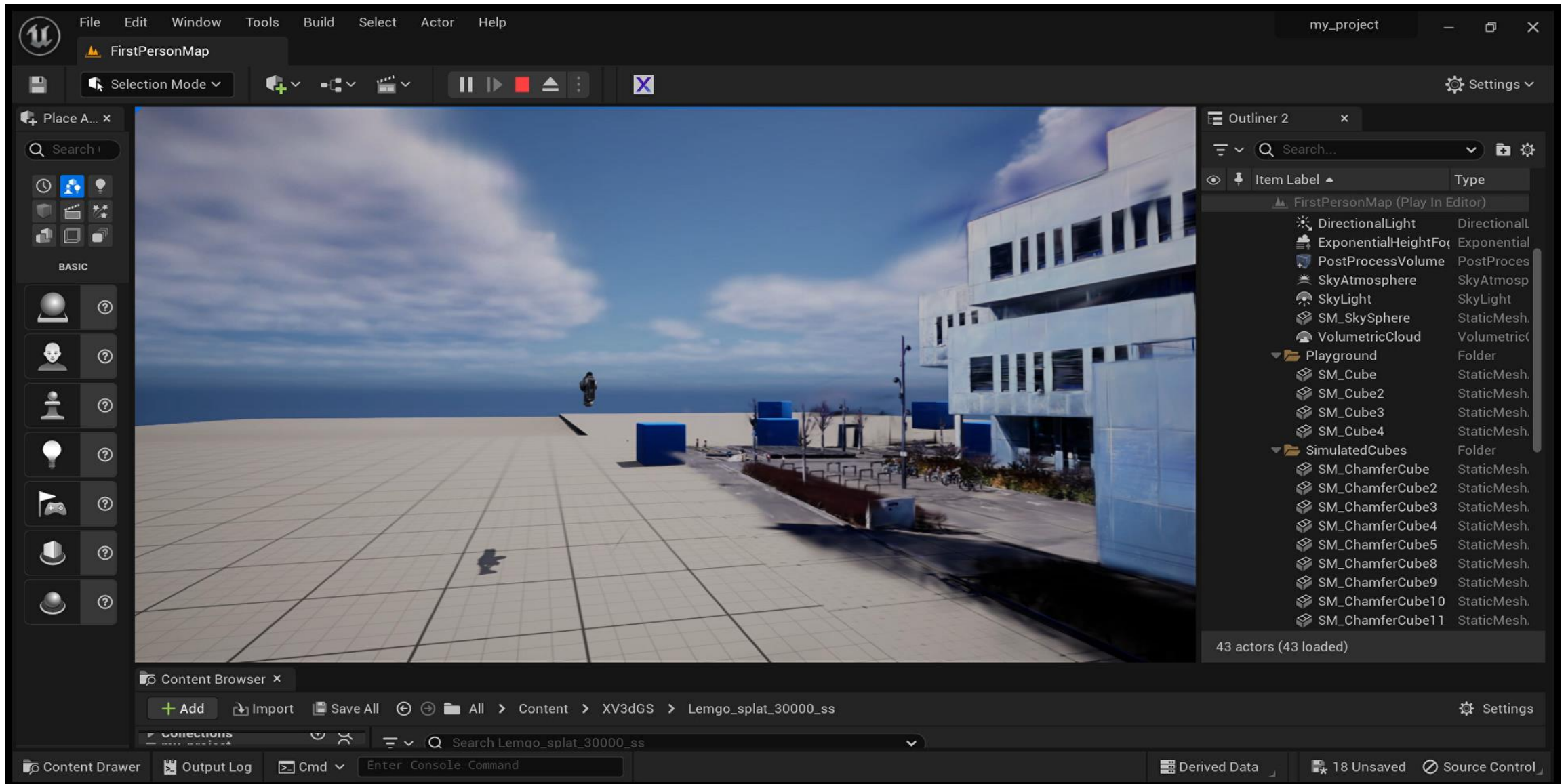


Unreal Engine:

- Imported the Gaussian Splat (.PLY) into Unreal Engine using X3DGS plugin
- Used First-Person template to enable interaction with the environment
- Added interactive features (movement +shooting + target blocks)



Unreal Engine Demo — Interactive Innovation SPIN



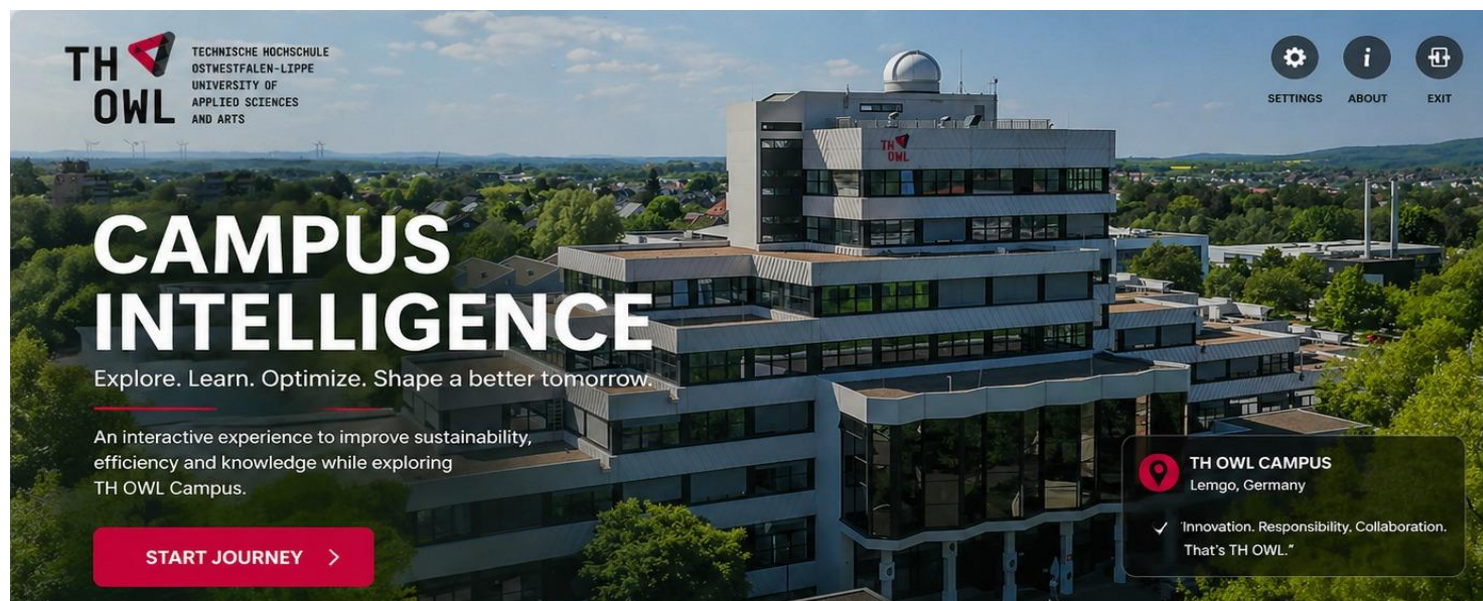
Limitations & Improvements

Limitations:

- Winter lighting caused shadows.
- Drone-based capture was not permitted, so we had to rely on slower manual GoPro acquisition
- Pipeline 1 required strong local GPU for stable high-quality output
- Pipeline 2 on DGX required reduced resolution due to memory and server availability limits

Improvements

- Capture in stable daylight.
- Use drone-based capture for faster and wider coverage
- Use cloud GPU platforms like Vast.ai and Lambda Labs, offering NVIDIA A100 (\$0.6–\$2/hr) and H100 (\$2–\$3/hr), to speed up Gaussian Splat model processing.



Future Scope - Gamified Campus Digital Twin

Transform the reconstructed TH OWL campus from a static Gaussian Splat model into an interactive **Campus Intelligence** platform where users can explore, learn, and optimise through gamified experiences.



SUSTAINABILITY – PLANT
TREES, PLACE SOLAR
PANELS, OPTIMISE ROUTES



EFFICIENCY – IMPROVE
ENERGY USE THROUGH
SMART BUILDING CONTROL



KNOWLEDGE – EXPLORE
HOTSPOTS, LABS, AND
ANSWER QUIZZES



EXPLORATION – FREE
NAVIGATION, GUIDED
TOURS, HIDDEN LOCATIONS

THANK YOU